TARA FRIEDRICH
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Open to opportunities in the SF Bay Area or remote

Programming Languages: Python, R, SQL

Data Science: Pandas, Numpy, Scikit-learn, PyTorch, Matplotlib, Ggplot2, Bioconductor, RShiny

Computational Resources: High Performance Cluster, AWS (S3, EC2, Batch, Lambda), Google Cloud Compute, Kubernetes, Docker, Argo, WDL, Nextflow, Prefect, DNAnexus, Git, JIRA, SQL (MySQL, Spark), noSQL (MongoDB, DynamoDB), Neo4j

Statistics and Machine Learning: Hypothesis Testing, Linear and Generalized Linear Modeling, Multivariate Modeling, Clustering, Dimensionality Reduction, Supervised and Unsupervised Classification, Random Forests, HMM, Kaplan-Meier

Summary

Scientist with 10+ years of experience integrating multimodal clinical and genomic data where statistics and machine learning are fundamental to building reliable products and solving real problems.

Experience

Bioinformatics Engineer - Metaphore Biotechnologies — March 2025–Present

· Scaled and refactored antibody screening and maturation pipelines using Nextflow to improve reproducibility, performance, and cross-campaign data mining for novel candiate binders.

· Designed and performed advanced analytics on LIMS data lake extracts to validate assay performance, identifying key experimental trends and ensuring data reliability for downstream modeling.

· Built a scalable pipeline to iteratively cluster roughly 2 billion antibody sequences by sequence similarity to accelerate identification of high-priority binders for downstream validation.

· Contributed to the design of knowledge graphs to advance research objectives, enabling scalable comparison of binders by sequence similarity, developability, and biophysical constraints across discovery campaigns.

· Delivered quarterly presentations on project progress and communicated broader technical and strategic insights in company-wide presentations.

Senior Bioinformatics Scientist - Natera — Feb 2022–August 2024

· Partnered with lab scientists to enhance RNA-seq assays, focusing on detecting oncogenic fusions in noisy FFPE tumor samples, leading to improved assay sensitivity and specificity.

· Designed novel quality metrics and visualization tools for germline and somatic pipelines, enabling biomarker-driven patient stratification and deeper characterization of assay performance.

· Built scalable, cloud-native pipelines that reduced turnaround time for WES and RNA-seq QC analyses by 40%.

· Developed an integrated classifier to identify pathogenic genetic variants by combining population genetics, functional annotations, machine learning-based predictors (e.g., AlphaFold), and curated clinical metadata.

Bioinformatics Scientist - Fabric Genomics — Nov 2019–Jan 2022

· Maintained and enhanced the ACE algorithm, a production-grade tool for prioritizing clinically pathogenic variants, by implementing comprehensive regression and unit testing to ensure reliability and compliance in a regulated environment.

· Enhanced variant prediction classifiers to increase precision in identifying likely pathogenic variants and flagging borderline cases for use by CAP/CLIA regulated labs using statistics and machine learning.

· Utilized clinical ontologies (HPO, SNOMED, ICD-9/10) to map patient symptoms to structured phenotypes and prioritize candidate variants for rare disease diagnosis and clinical interpretation.

· Leveraged distributed computing frameworks like Apache Spark to scale data classification workflows and optimize ETL processes, streamlining data analysis pipelines for large-scale datasets.

· Actively participated in code reviews, planning sessions, and best practices discussions, fostering a collaborative and high-quality development environment.

Education

Ph.D. Bioinformatics - University of California, San Francisco

· Statistics and machine learning to identify enhancers and genes predictive of tissue developmental phenotypes.

B.S. Biochemistry - University of California, Los Angeles

linkedin.com/in/tara-friedrich-bioinformatics · github.com/staracode

Publications

Cordero, G.A., Holloway, A.K., **Friedrich, T.**, Eme, J., Eckalbar, W., Kusumi, K., Janzen, F.J., Hicks, J.W., Conlon, F.L., Bruneau, B.G., and Pollard, K.S. (2025). **The Interplay of Ontogeny and Phylogeny at the Transcriptome Level of the Tetrapod Heart.** *Journal of Molecular and Developmental Evolution*.

Friedrich, T.*, Betty, B.*, Mason, M., VanderMeer, J., Zhao, J., Eckalbar, W., Logan, M., Pollard, K.S., Illing, N., and Ahituv, N. (2016). **Bat accelerated regions identify a bat forelimb specific enhancer in the HoxD locus.** *PLoS Genetics*, 12(3):e1005738. (*Co-first authors).

Eckalbar, W., Schlebusch, S., Mason, M., Gill, Z., Booker, B., Nishizaki, S., Nday, C., Terhune, E., Nevonen, K., Makki, N., **Friedrich, T.**, VanderMeer, J., Pollard, K.S., Carbone, L., Wall, J., Illing, N., Parker, A., and Ahituv, N. (2016). **Genomic characterization of the developing bat wing.** *Nature Genetics*, 48, 528–536.

Myers, S.A., Petted, S., Chatterjee, N., **Friedrich, T.**, Tomoda, K., Krings, G., Thomas, S., Broeker, M., Maynard, J., Thomson, M., Pollard, K., Yamanaka, S., Burlingame, A.L., and Panning, B. (2016). **SOX2 O-GlcNAcylation alters its protein-protein interactions and genomic occupancy to modulate gene expression in pluripotent cells.** *eLife*, 5:e10647.

Kostka, D., **Friedrich, T.**, Holloway, A.K., and Pollard, K.S. (2015). **motifDiverge: a model for assessing the statistical significance of gene regulatory motif divergence between two DNA sequences.** *Statistics and Its Interface*, 8(4), 463–476.

Fogel, B.L., Wexler, E., Wahnich, A., **Friedrich, T.**, Vijayendran, C., Gao, F., Parikshak, N., Konopka, G., and Geschwind, D.H. (2012). **RBFOX1 Regulates Both Splicing and Transcriptional Networks in Human Neuronal Development.** *Human Molecular Genetics*, 21(18), 4171–4186.

Konopka, G., **Friedrich, T.**, Davis-Turak, J., Winden, K., Oldham, M.C., Gao, F., Chen, L., Wang, G., Luo, R., Preuss, T.M., and Geschwind, D.H. (2012). **Human-Specific Transcriptional Networks in the Brain.** *Neuron*, 75, 601–617.

Schein, S., Sands-Kidner, M., and **Friedrich, T.** (2008). **The physical basis for the head-to-tail rule that excludes most fullerene cages from self-assembly.** *Biophysical Journal*, 94, 938–957.

Schein, S. and **Friedrich, T.** (2008). **A geometric constraint, the head-to-tail exclusion rule, may be the basis for the isolated-pentagon rule in fullerenes with more than 60 vertices.** *Proceedings of the National Academy of Sciences*, 105, 19142–19147.

Nagarajan, S., **Friedrich, T.**, Garcia, M., Kambham, N., and Sarwal, M.M. (2005). **Gastrointestinal leukocytoclastic vasculitis: an adverse effect of sirolimus.** *Pediatric Transplantation*, 9, 97–100.